

Sajil Lidhan

Generative AI Backend Engineer | Java Backend Developer

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[LinkedIn](#) | [Portfolio](#)

Professional Summary

Generative AI Backend Engineer with 4 years of experience in Java backend development, enterprise-scale applications, and automation solutions for telecom and analytics platforms. Proficient in Java, Spring Boot, Microservices, REST APIs, Python, Kafka, SQL, Oracle, MySQL, Docker, Kubernetes, and Google Cloud Platform, with expertise in cloud-native development, distributed systems, and workflow automation. Experienced in applying Generative AI, prompt engineering, and AI-assisted development to accelerate software delivery, improve code quality, and enhance engineering productivity. Passionate about building scalable, high-performance, AI-powered enterprise applications that deliver measurable business value.

Skills

Programming Languages:	Java, Python, SQL, Bash/Shell
Backend Technologies:	Spring Boot, Microservices, REST APIs, API Integration
AI & Generative AI:	Generative AI, Prompt Engineering, AI-Assisted Development, Natural Language Processing (NLP), AI Agents, Model Context Protocol (MCP)
Cloud & DevOps:	Docker, Kubernetes, Google Cloud Platform (GCP), CI/CD
Messaging & Streaming:	Apache Kafka, Event-Driven Architecture
Automation & Integration:	Workflow Automation, ETL, Pentaho, n8n
Developer Tools:	Git, Maven, Postman, Jira
Architecture:	Distributed Systems, Cloud-Native Development, Scalable Backend Systems
Core Concepts:	Data Structures & Algorithms, Performance Optimization, Agile (Scrum), SDLC

Experience

SOFTWARE DEVELOPER | Generative AI & Backend Engineering | 6D Technologies (Telecom Product Company) *August 2022 – Present | Bangalore*

Backend Engineering & Enterprise Automation

- Architected and implemented scalable JSON transformation workflows from complex telecom business requirements, improving processing efficiency by 30% while enabling faster onboarding of enterprise use cases.
- Developed a configurable JSON-to-XML transformation framework with dynamic profile-to-column mapping across Oracle, MySQL, Hive, and HBase, increasing system flexibility by 40% and streamlining cross-project integration.
- Engineered automated SQL generation and ETL workflows using Pentaho KTR and Redisson caching, reducing processing latency by 35% and eliminating over 10 hours of manual effort per week.
- Developed enterprise-grade backend services using Java, Spring Boot, REST APIs, and Microservices, supporting scalable and high-performance telecom applications.

Data Processing & Integration

- Designed and deployed a Python-based customer segmentation automation solution, processing large-scale datasets with 60% faster execution and 98% accuracy, improving operational scalability.
- Integrated external promotion and rewards platforms through REST APIs and Apache Kafka event streams, ensuring reliable high-volume transaction processing and seamless third-party connectivity.
- Built workflow automation solutions supporting large-scale data transformation, system integration, and business process optimization across enterprise environments.

Generative AI & Intelligent Automation

- Contributed to the development of Aarya, an AI-powered assistant enabling campaign creation, text-to-rule generation, and virtual profile configuration through intelligent automation and natural-language-driven workflows.
- Developed automation capabilities that transformed natural-language business requirements into structured JSON rule definitions, accelerating rule creation and reducing manual configuration effort.

- Applied Generative AI, prompt engineering, and AI-assisted development practices to accelerate feature delivery, improve debugging efficiency, automate documentation, and enhance engineering productivity.

Cloud & Platform Engineering

- Containerized applications using Docker and Kubernetes and supported CI/CD deployments on Google Cloud Platform (GCP), doubling deployment velocity while maintaining 99% production availability.
- Supported deployment automation and environment management across cloud-native platforms, improving release consistency and operational reliability.

Collaboration & Technical Documentation

- Collaborated with product managers, business analysts, QA teams, and client stakeholders to translate complex telecom business requirements into scalable technical solutions.
- Participated in end-to-end software development activities including requirement analysis, solution design, implementation, testing, deployment, and production support.
- Authored technical documentation, deployment guides, and knowledge-sharing materials to improve team productivity and support efficient onboarding.
- Contributed to code reviews, troubleshooting activities, and cross-functional discussions to ensure high-quality software delivery and operational reliability.

Key Projects

AI-Powered Conversational Assistant using Amazon Lex | React Native | AWS | GitHub Repository

Developed a cross-platform conversational assistant using Amazon Lex and React Native, enabling users to interact with intelligent chatbot workflows through natural-language conversations.

- Built a mobile chatbot application using React Native and Amazon Lex to deliver conversational user experiences across multiple use cases.
- Integrated Amazon DynamoDB and backend services to manage user interactions, session data, and chatbot workflows.
- Designed and configured intent-based conversation flows, improving user engagement and automating common user interactions.
- Implemented scalable cloud-based chatbot architecture using AWS services to support real-time conversational experiences.
- **Technologies:** React Native, Amazon Lex, AWS, DynamoDB, REST APIs, Conversational AI.

Natural Language to SQL Analytics Assistant | Python | NLP | n8n | GitHub Repository

Developed an intelligent analytics assistant capable of converting natural-language business queries into SQL statements and presenting results through conversational responses, charts, and tabular visualizations.

- Built a text-to-SQL workflow that transformed user questions into executable database queries, simplifying data access for non-technical users.
- Enabled dynamic result visualization through chat responses, graphical reports, and tabular dashboards.
- Integrated Telegram and n8n workflows to support conversational analytics and automated query execution.
- Improved accessibility to enterprise data by reducing manual SQL creation and streamlining business reporting workflows.
- **Technologies:** Python, SQL, Telegram Bot API, n8n, Data Visualization, Workflow Automation, Natural Language Processing (NLP)

Certifications

- Generative AI Fundamentals – Microsoft
- Google Cloud Skills Boost – Google Cloud
- AWS Certified Cloud Practitioner (CLF-C02)

Education

Master of Computer Applications (MCA) January 2022

APJ Abdul Kalam Technological University, Thiruvananthapuram, India

Bachelor of Computer Applications (BCA) March 2019

University of Calicut, Kerala, India